Perfect Network Resilience in Polynomial Time

Matthias Bentert Stefan Schmid

Technische Universität Berlin

Salt Lake City, June 23 2026

STOC

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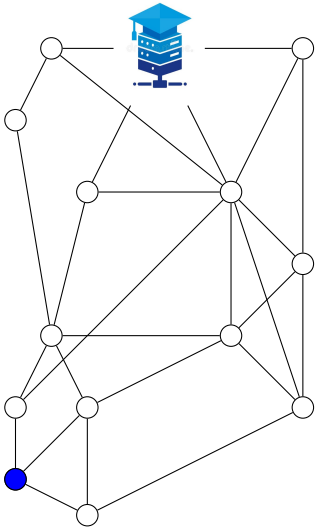

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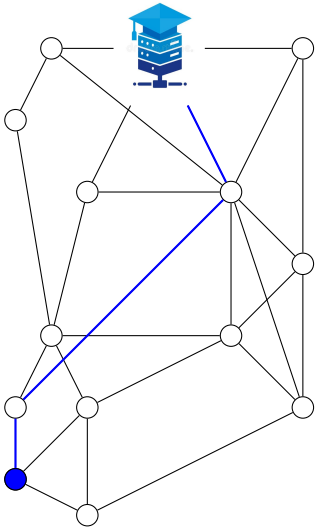

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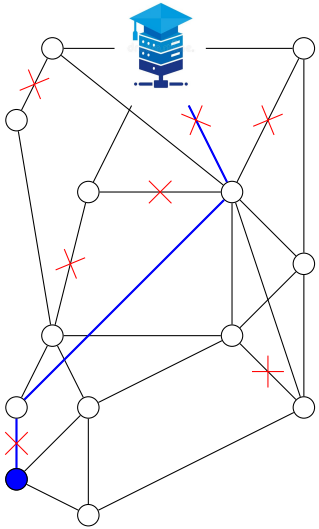

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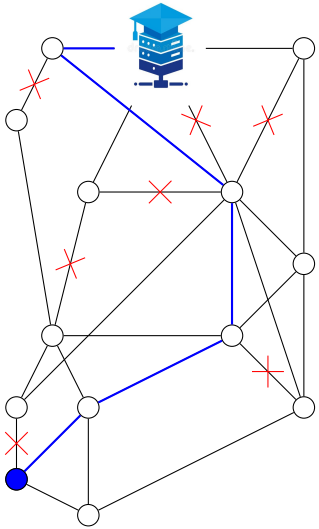

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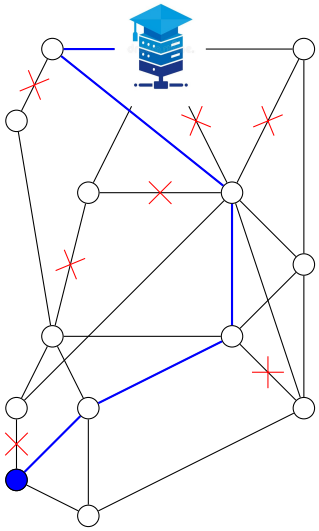

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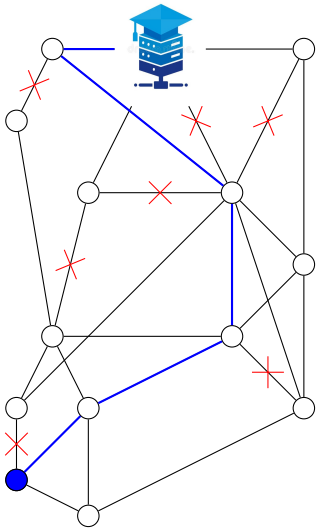

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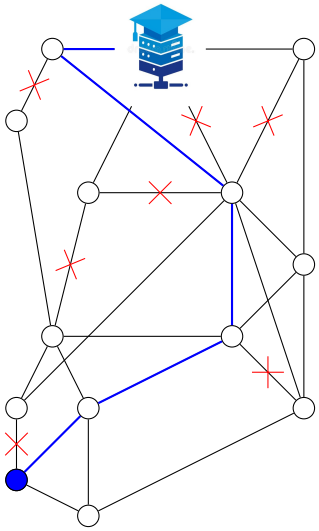

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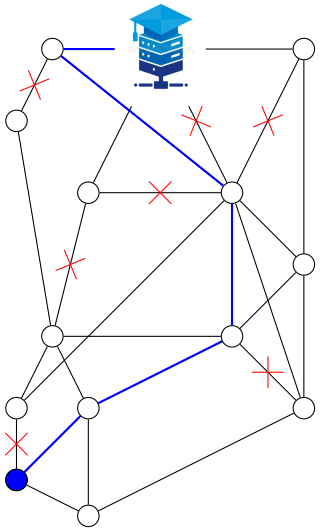

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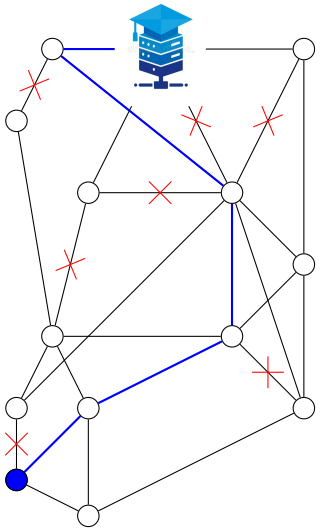

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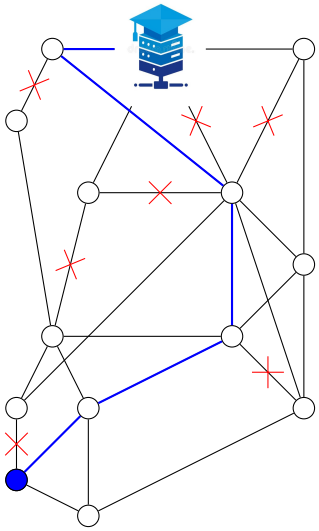

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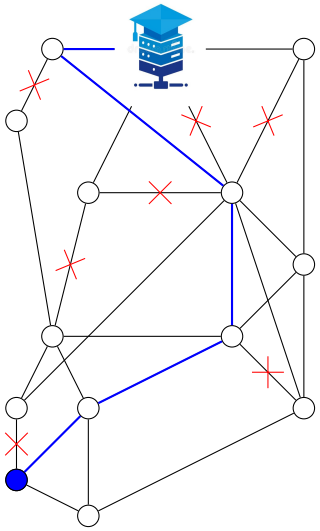

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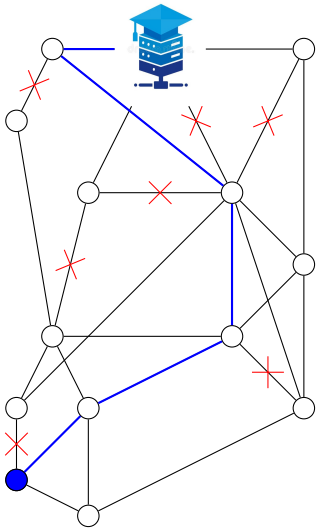

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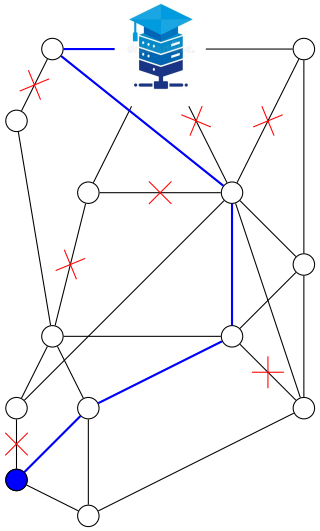

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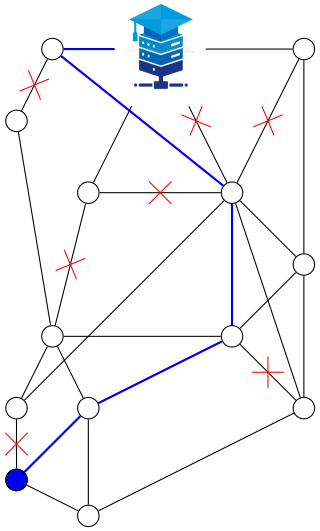

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Input: A graph G and a vertex t.
Question: Do perfectly resilient forwarding rules exist?

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Perfect Network Resilience Synthesis

Input: A graph G and a vertex t.
Task: Design perfectly resilient forwarding rules for each vertex or determine that none exist.

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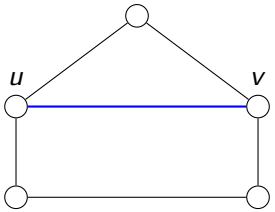

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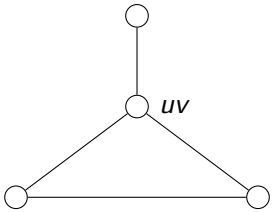

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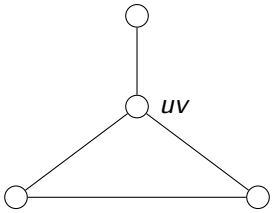

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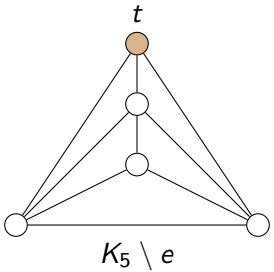

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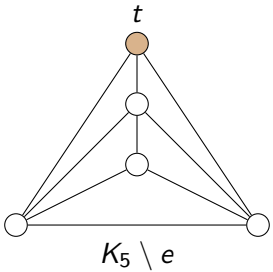

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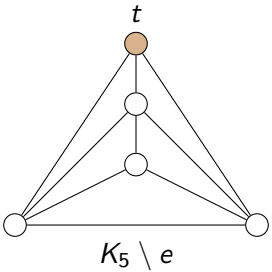

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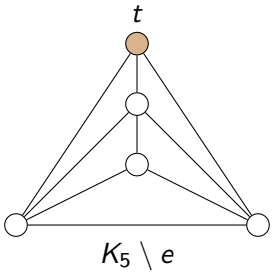

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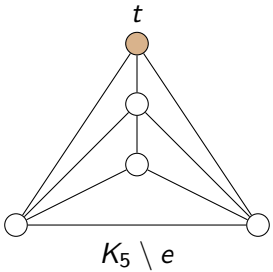

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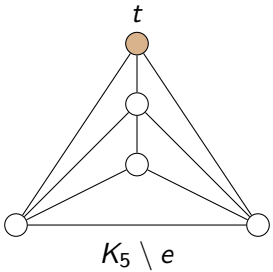
- Compact encoding of solutions powerful enough? [Chiesa, Nikolaevskiy, Mitrovic, Gurtov, Madry, Schapira, Shenker. ToN '17] [Foerster, Hirvonen, Pignolet, Schmid, Trédan. APOCS '21]



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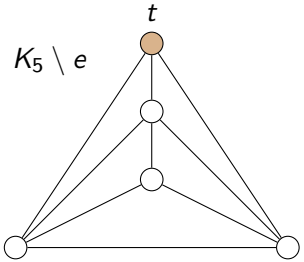
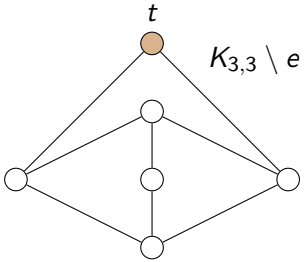
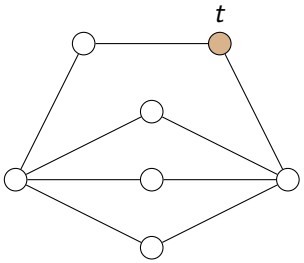
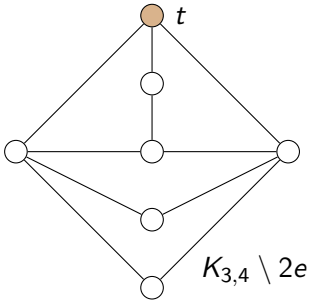






subdivided $K_{2,4}$



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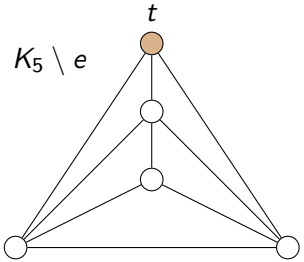

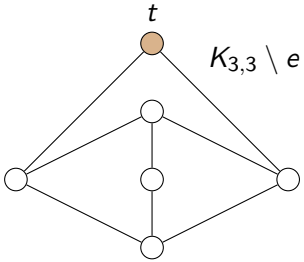

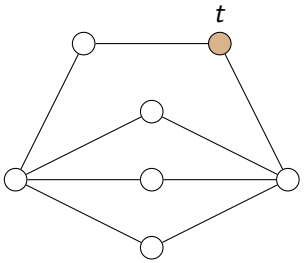

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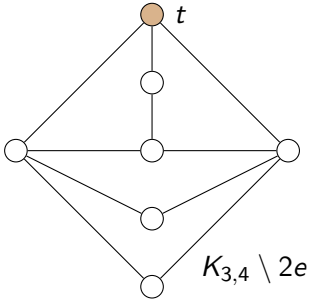

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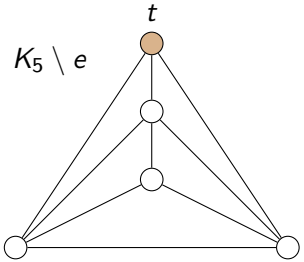

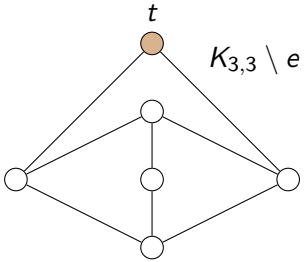

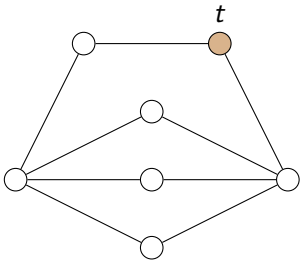

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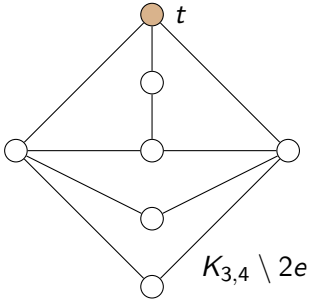

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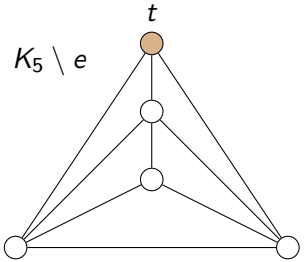

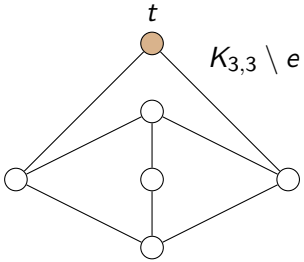

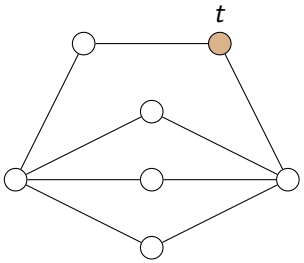

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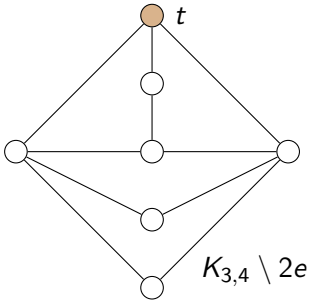

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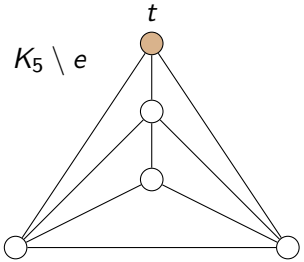

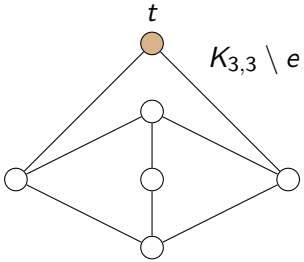

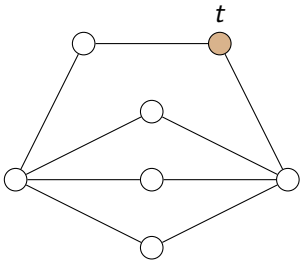

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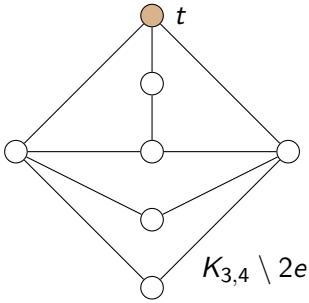

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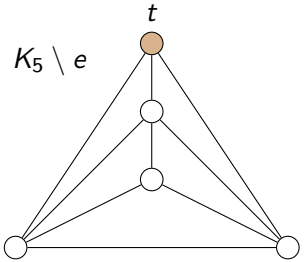

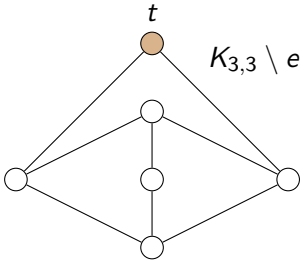

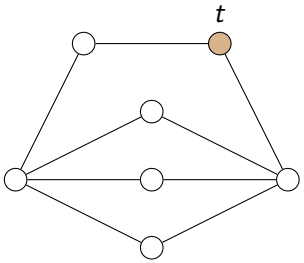

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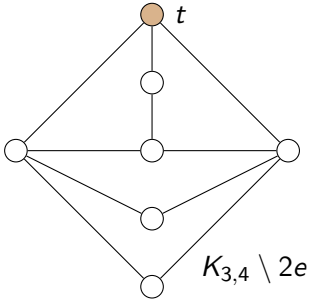

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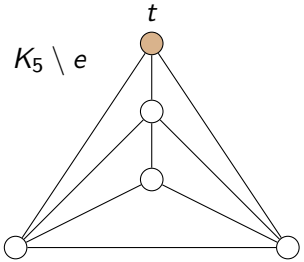

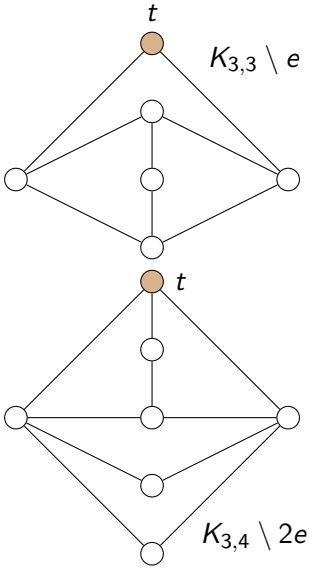

High-Level Overview I

Structural Analysis

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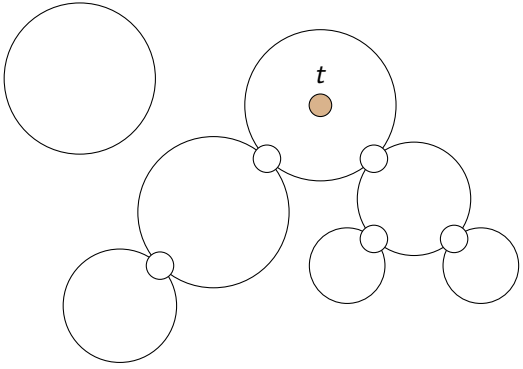

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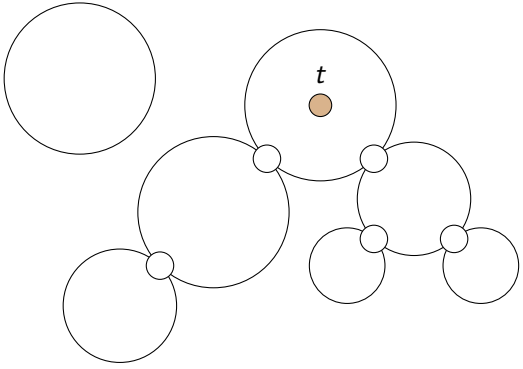

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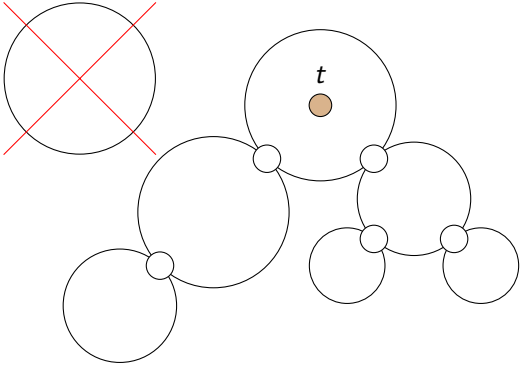

Preprocessing

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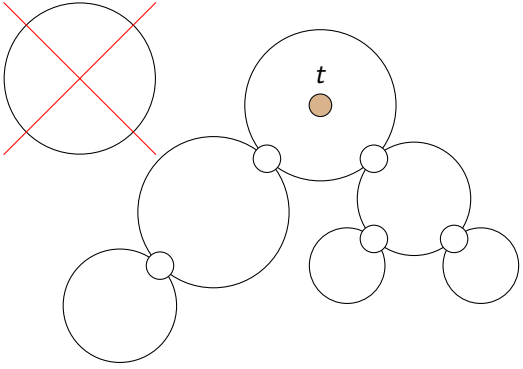

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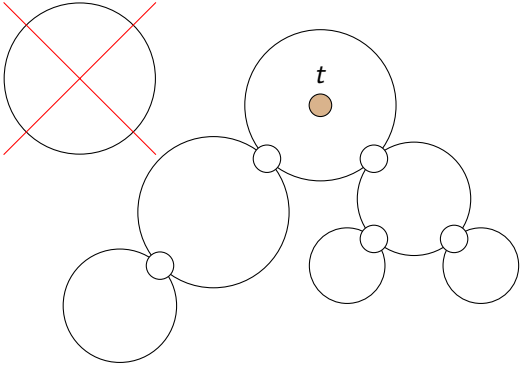

Theorem (Wagner. Math. Ann. '37)

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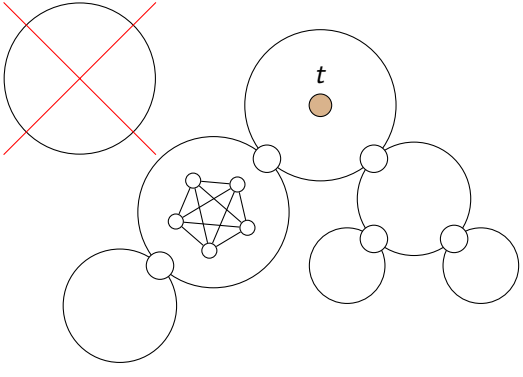

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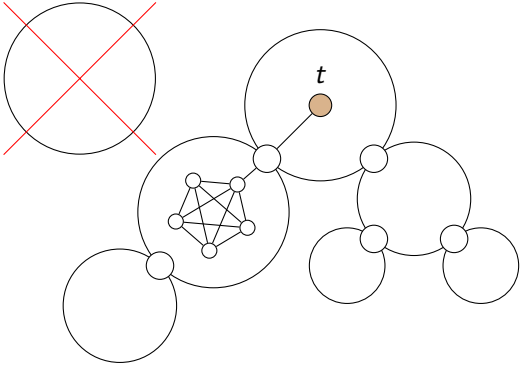

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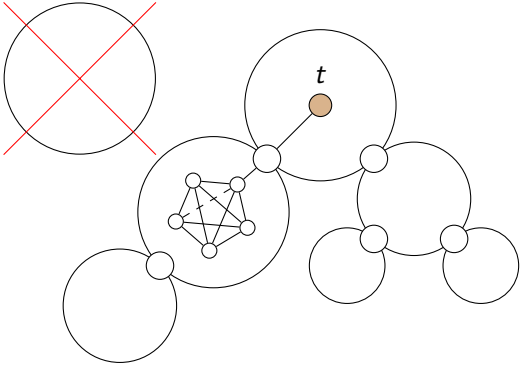

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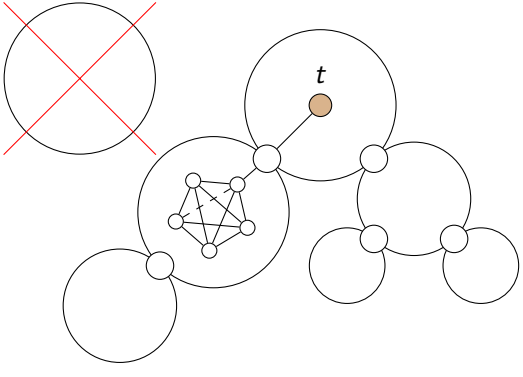

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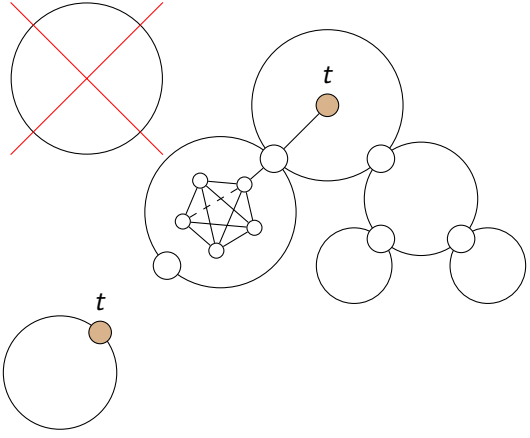

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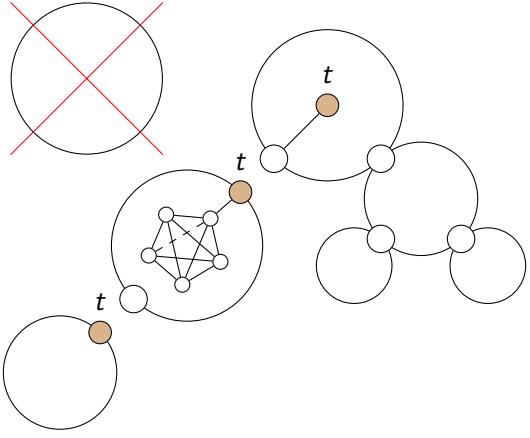

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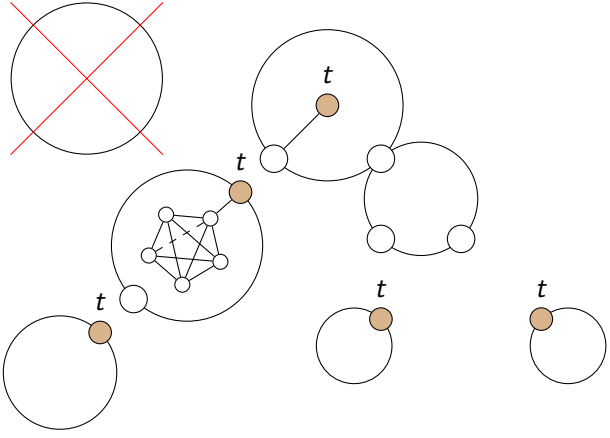

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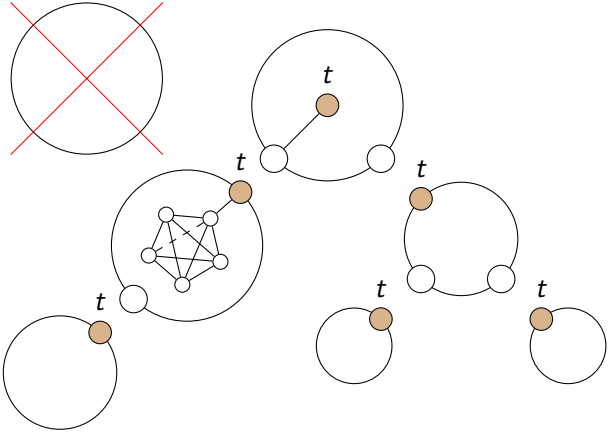

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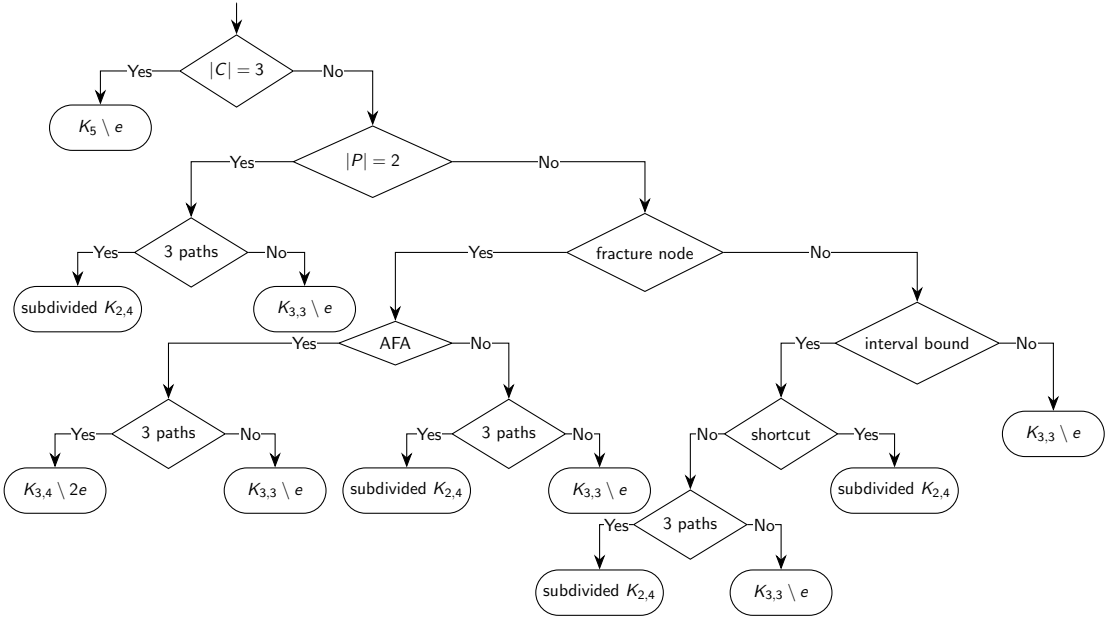

Linear-Time Algorithm for Decision

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- Check whether connected component C containing t is planar
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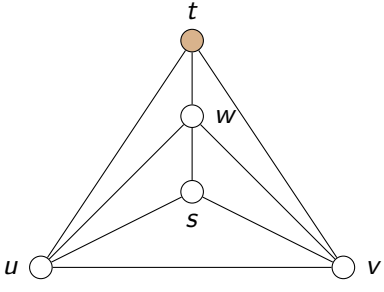
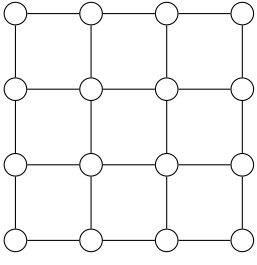
$\rightsquigarrow$ Connected component of t has branchwidth at most 9 or contains $4x4$-grid as minor

Grids





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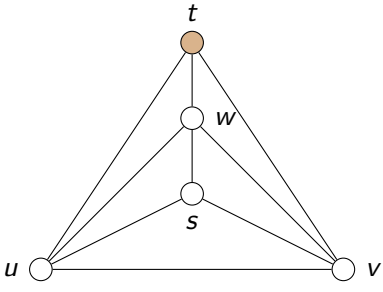
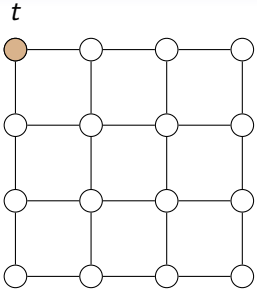
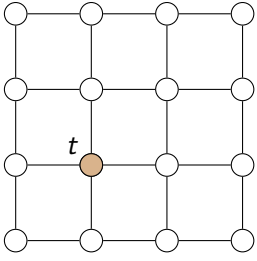
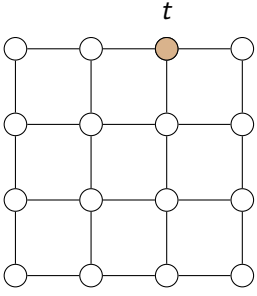
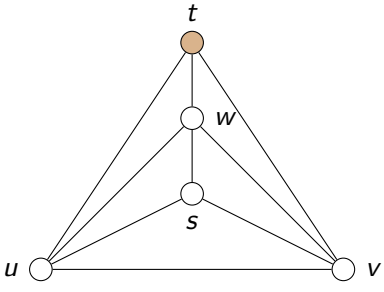


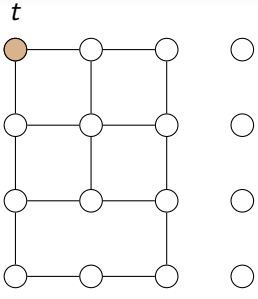


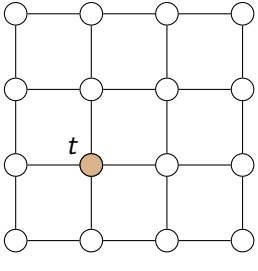


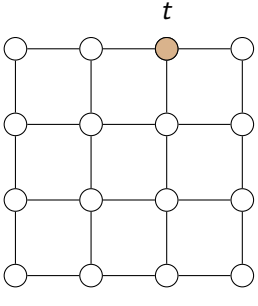


Grids









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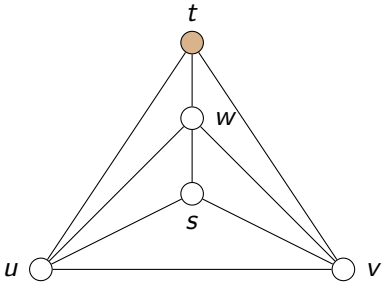

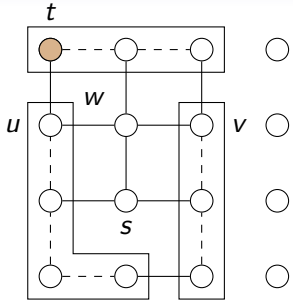

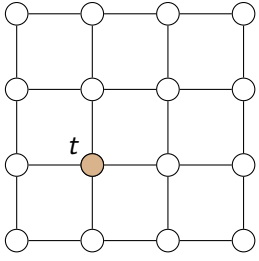

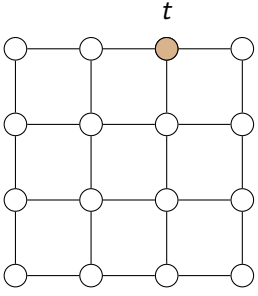

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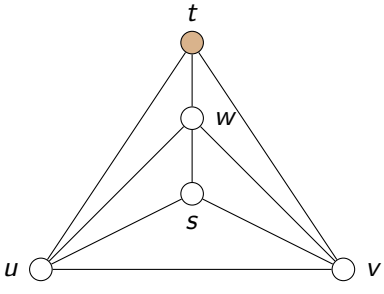

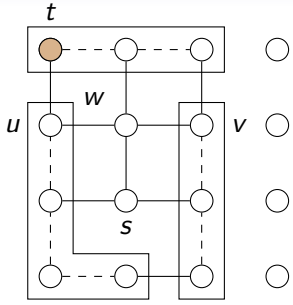

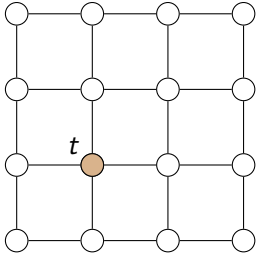

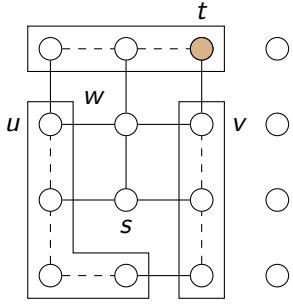

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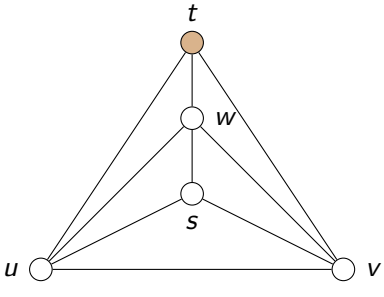
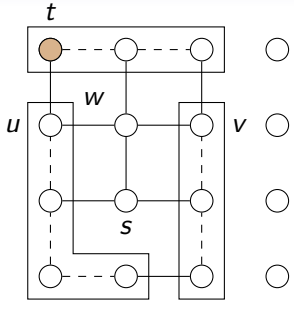
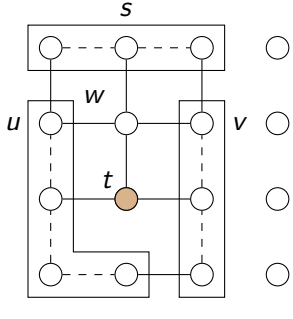
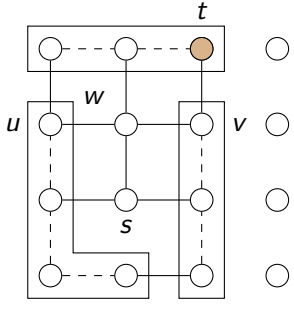


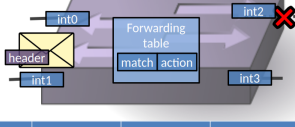






Future Work

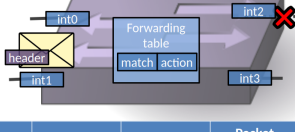
Future Work



Per-destination	Per source	Incoming port	Packet header rewriting

Future Work

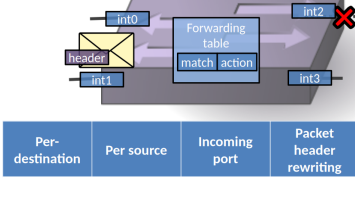
1. Additional Information
   - Source vertex



Per-destination	Per source	Incoming port	Packet header rewriting

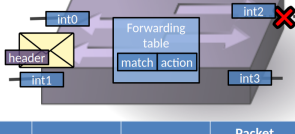
Future Work

1. Additional Information
   - Source vertex
   - Modifiable header bits



Future Work

1. Additional Information
   - Source vertex
   - Modifiable header bits
2. Different Failure Models
   - (Semi-)dynamic edge failures



Future Work

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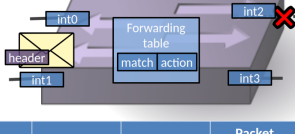
2. Different Failure Models
   - (Semi-)dynamic edge failures
   - Vertex failures



Per-destination	Per source	Incoming port	Packet header rewriting

Future Work

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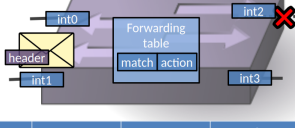
   - (Semi-)dynamic edge failures
   - Vertex failures
3. Weaker Guarantee
   - Ideal resilience (any k-connected graph is $(k-1)$-resilient)



Per-destination	Per source	Incoming port	Packet header rewriting

Future Work

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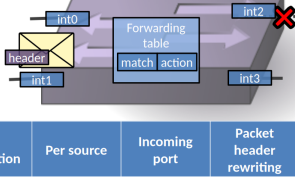
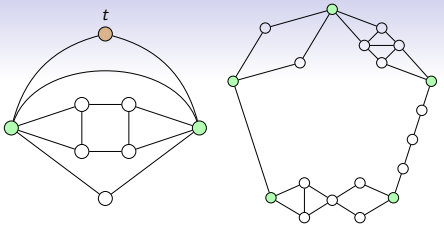
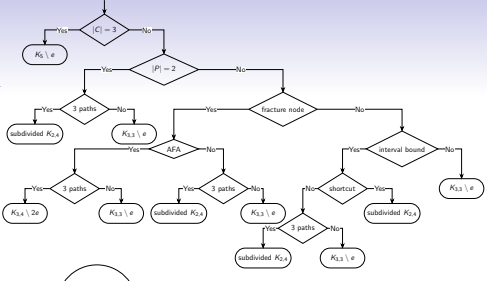
Thank you.





Future Work



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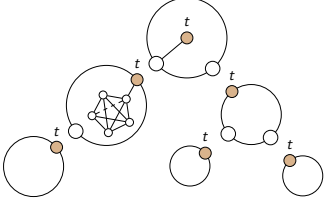
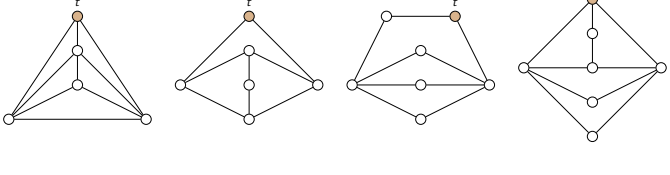
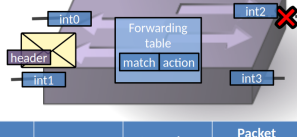


Thank you.





Per-destination	Per source	Incoming port	Packet header rewriting